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(54) **DOMAIN GENERALIZATION FOR MACHINE LEARNING MODELS**

(71) Applicant: **INTERNATIONAL BUSINESS MACHINES CORPORATION**, Armonk, NY (US)

(72) Inventors: **Shaked Perek**, Givatayim (IL); **Amir Egozi**, Eliav (IL); **Efrat Hexter**, Beit Shemesh (IL); **Assaf Arbelle**, Lehvot Haviva (IL)

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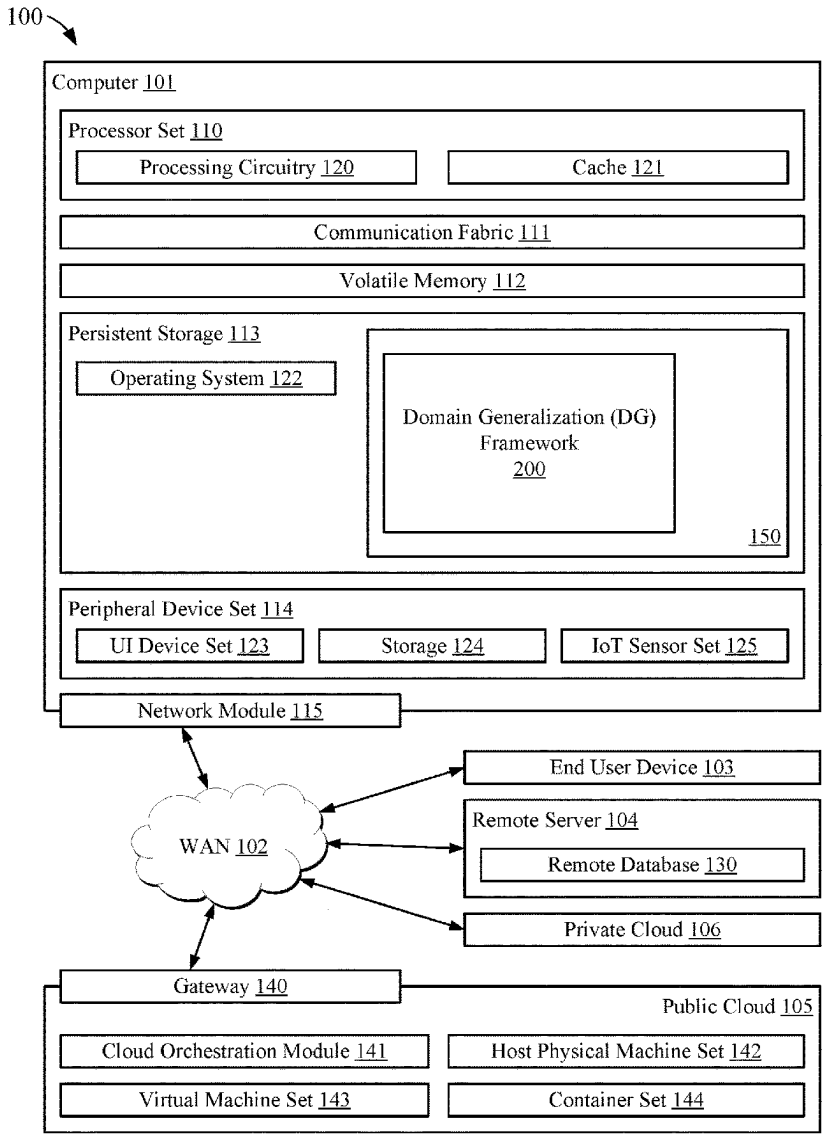
(57) **ABSTRACT**

Training a machine learning model for domain generalized operation includes processing, using computer hardware, a first plurality of images belonging to a first domain through a first network. A second plurality of images belonging to a second domain is processed using the computer hardware through a second network. A compound error metric is generated using the computer hardware from a plurality of plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network. Weights of the first network are updated using the computer hardware based on the compound error metric. Weights of the second network are updated using the computer hardware using a moving average technique that is dependent on the weights of the first network as updated.



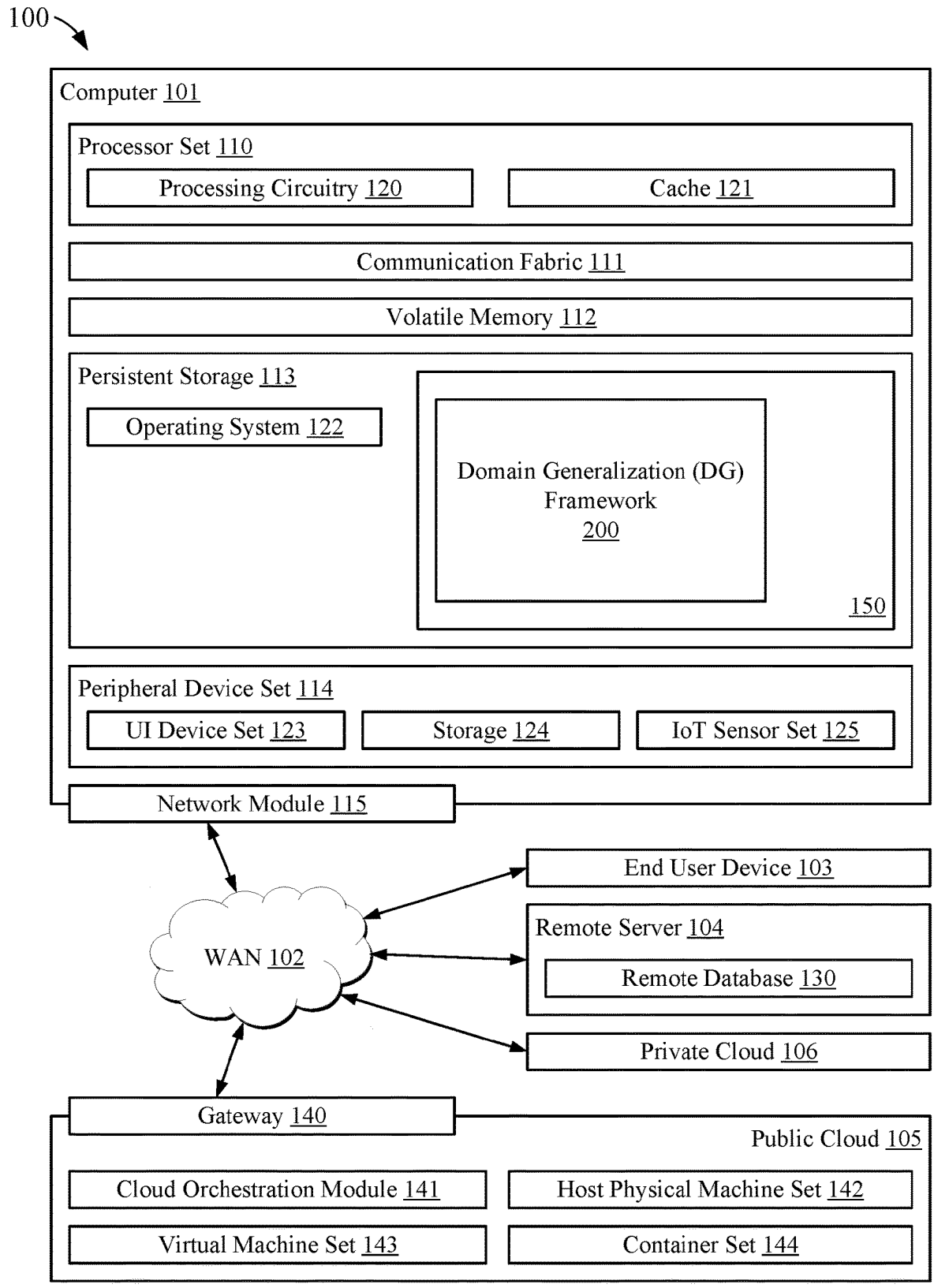

100
Computer 101
Processor Set 110
Processing Circuitry 120
Cache 121
Communication Fabric 111
Volatile Memory 112
Persistent Storage 113
Operating System 122
Domain Generalization (DG) Framework 200
150
Peripheral Device Set 114
UI Device Set 123
Storage 124
IoT Sensor Set 125
Network Module 115
WAN 102
End User Device 103
Remote Server 104
Remote Database 130
Private Cloud 106
Gateway 140
Public Cloud 105
Cloud Orchestration Module 141
Host Physical Machine Set 142
Virtual Machine Set 143
Container Set 144

FIG. 1

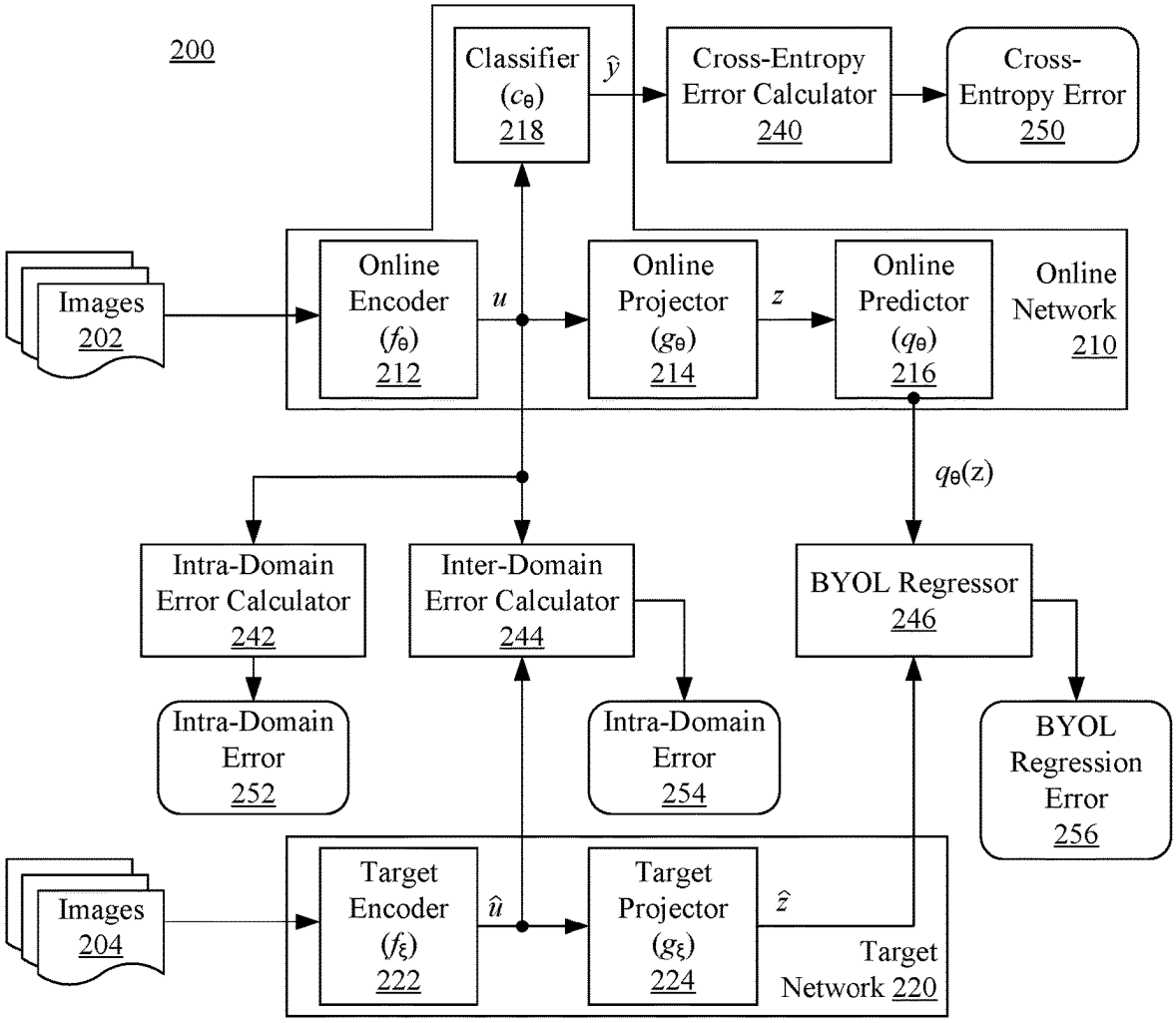

200
Classifier
(c_θ)
218
$\hat{y}$
Cross-Entropy
Error Calculator
240
Cross-
Entropy Error
250
Images
202
Online
Encoder
(f_θ)
212
u
Online
Projector
(g_θ)
214
z
Online
Predictor
(q_θ)
216
Online
Network
210
$q_\theta(z)$
Intra-Domain
Error Calculator
242
Inter-Domain
Error Calculator
244
BYOL Regressor
246
Intra-Domain
Error
252
Intra-Domain
Error
254
BYOL
Regression
Error
256
Images
204
Target
Encoder
(f_ξ)
222
$\hat{u}$
Target
Projector
(g_ξ)
224
$\hat{z}$
Target
Network 220

FIG. 2A

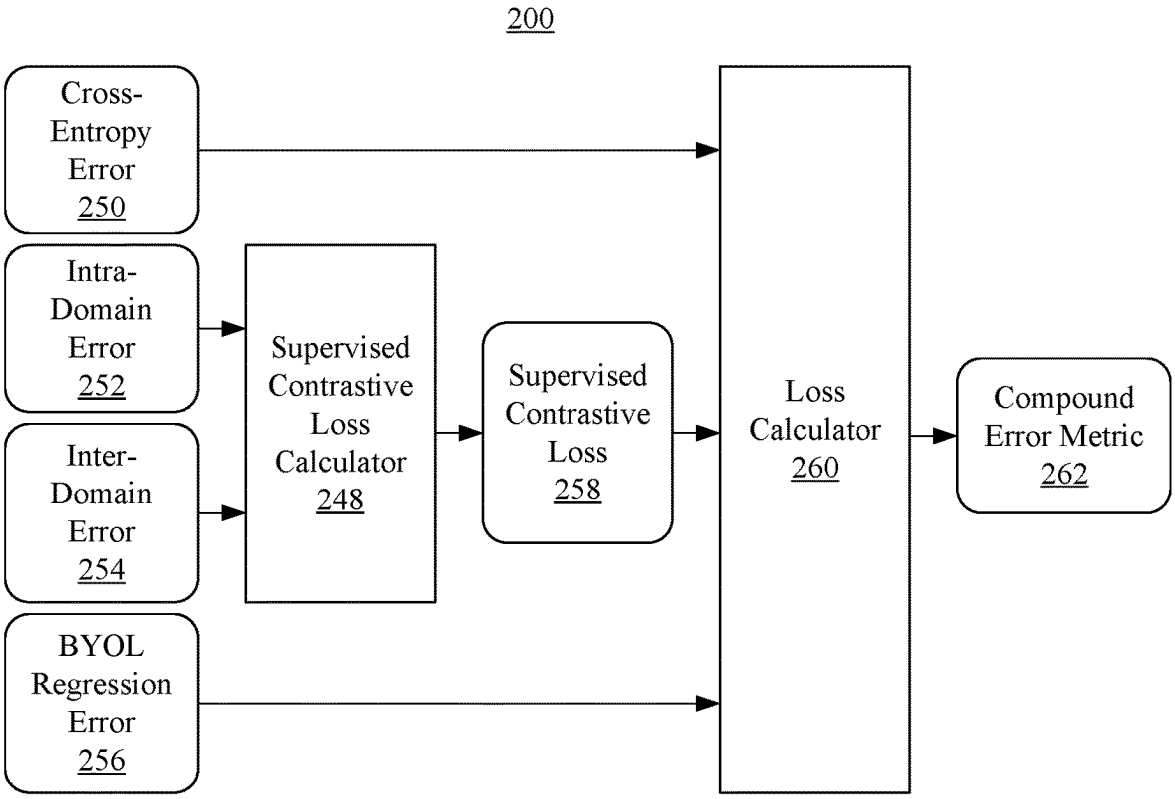

200
Cross-Entropy Error 250
Intra-Domain Error 252
Inter-Domain Error 254
BYOL Regression Error 256
Supervised Contrastive Loss Calculator 248
Supervised Contrastive Loss 258
Loss Calculator 260
Compound Error Metric 262

FIG. 2B

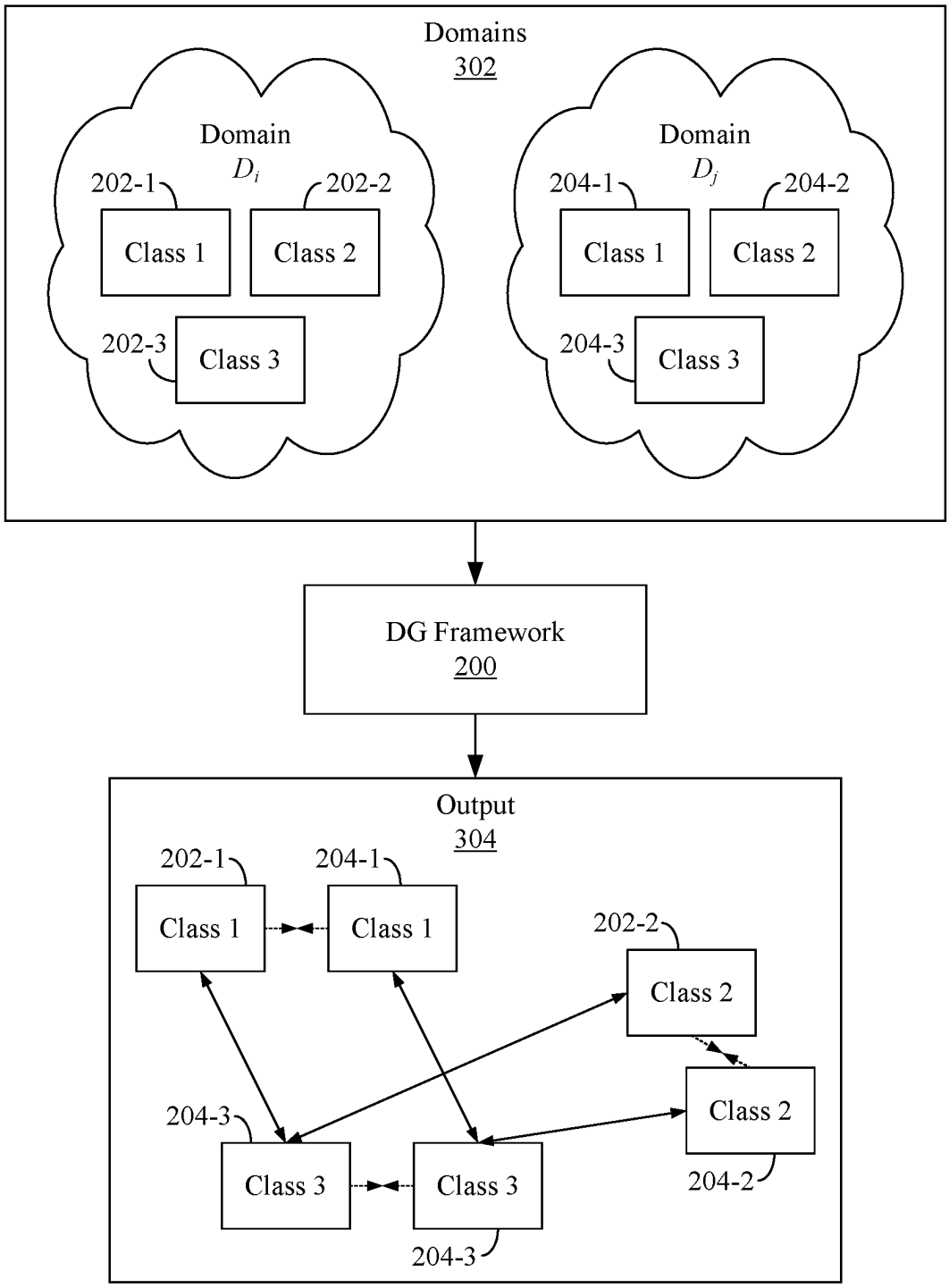

Domains
302
Domain
D_i
202-1
202-2
Class 1
Class 2
202-3
Class 3
Domain
D_j
204-1
204-2
Class 1
Class 2
204-3
Class 3
DG Framework
200
Output
304
202-1
204-1
Class 1
Class 1
202-2
Class 2
Class 2
204-3
Class 3
Class 3
204-2
204-3

FIG. 3

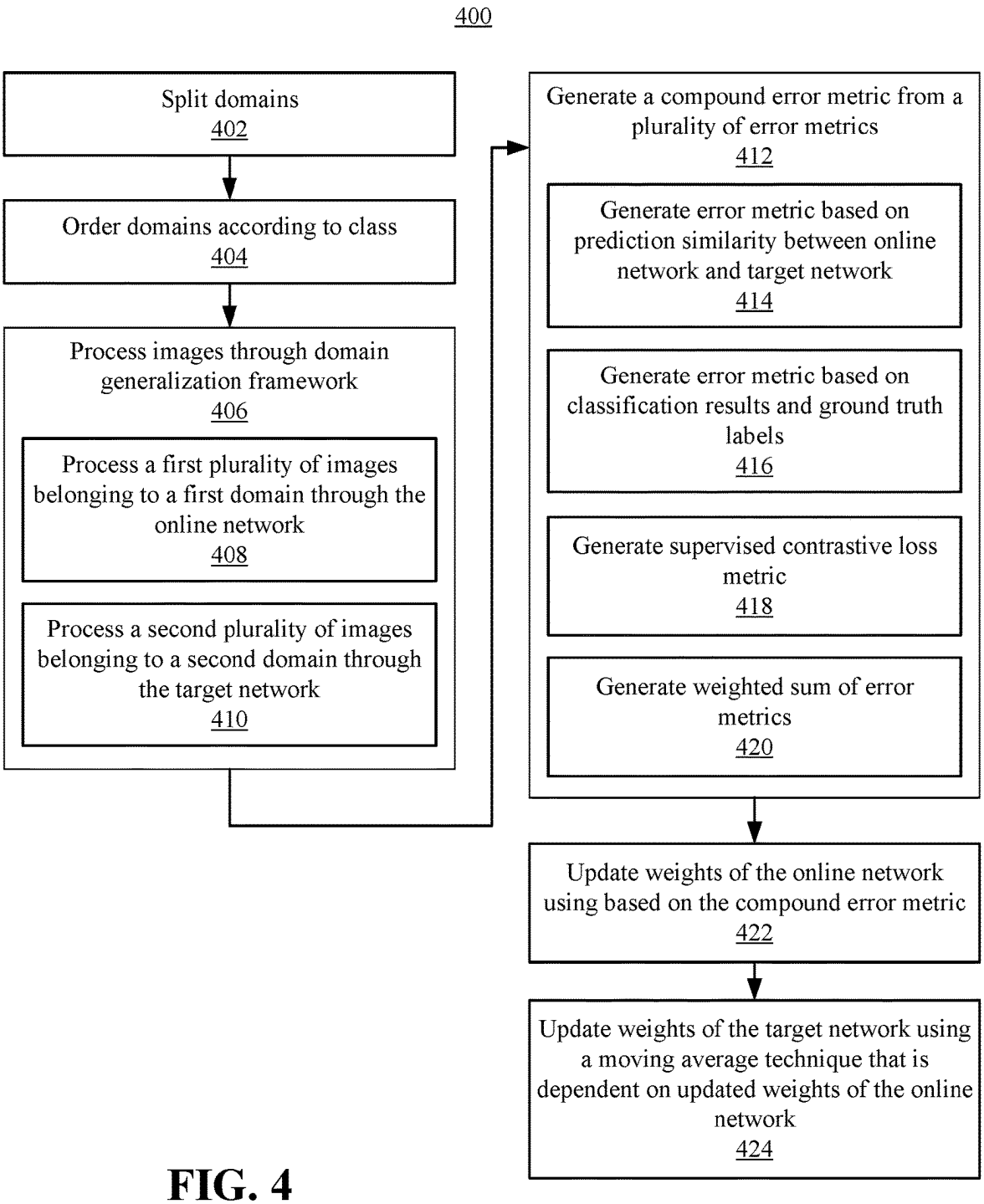

400
Split domains
402
Order domains according to class
404
Process images through domain generalization framework
406
Process a first plurality of images belonging to a first domain through the online network
408
Process a second plurality of images belonging to a second domain through the target network
410
Generate a compound error metric from a plurality of error metrics
412
Generate error metric based on prediction similarity between online network and target network
414
Generate error metric based on classification results and ground truth labels
416
Generate supervised contrastive loss metric
418
Generate weighted sum of error metrics
420
Update weights of the online network using based on the compound error metric
422
Update weights of the target network using a moving average technique that is dependent on updated weights of the online network
424

FIG. 4

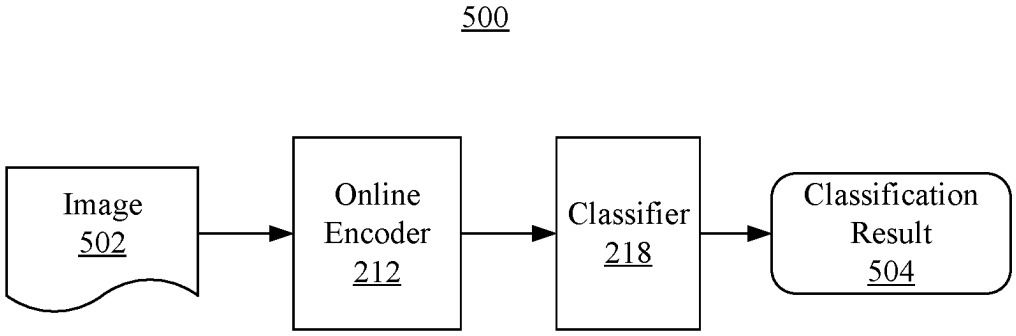

500
Image
502
Online
Encoder
212
Classifier
218
Classification
Result
504

FIG. 5

DOMAIN GENERALIZATION FOR MACHINE LEARNING MODELS

BACKGROUND

[0001] This disclosure relates to domain generalization for machine learning (ML) models.

[0002] ML models are usually trained and evaluated on datasets drawn from the same distribution. In other words, an ML model, such as a Deep Learning model, generally operates under the assumption that the data to be processed at runtime, or inference time, has a similar distribution as the training data used to train the ML model. This assumption holds true for a variety of different curated benchmark data sets such as the well-known ImageNet data set. In real-world scenarios, however, this assumption rarely holds true. In real-world scenarios, a shift in the data distribution between training time and inference time often causes a significant degradation in performance of the ML model.

SUMMARY

[0003] In one or more embodiments, a method of training a machine learning model includes processing, using computer hardware, a first plurality of images belonging to a first domain through a first network. The method includes processing, using the computer hardware, a second plurality of images belonging to a second domain through a second network. The method includes generating, using the computer hardware, a compound error metric from a plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network. The method includes updating, using the computer hardware, weights of the first network based on the compound error metric. The method includes updating, using the computer hardware, weights of the second network using a moving average technique that is dependent on the weights of the first network as updated.

[0004] In one or more embodiments, a system includes one or more processors configured to perform executable operations for training a machine learning model. The executable operations include processing a first plurality of images belonging to a first domain through a first network. The executable operations include processing a second plurality of images belonging to a second domain through a second network. The executable operations include generating a compound error metric from a plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network. The executable operations include updating weights of the first network based on the compound error metric. The executable operations include updating weights of the second network using a moving average technique that is dependent on the weights of the first network as updated.

[0005] In one or more embodiments, a computer program product includes a computer readable storage medium having program instructions stored thereon. The program instructions are executable by one or more processors to cause the one or more processors to execute operations for training a machine learning model. The executable operations include processing a first plurality of images belonging to a first domain through a first network. The executable operations include processing a second plurality of images belonging to a second domain through a second network. The executable operations include generating a compound error metric from a plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network. The executable operations include updating weights of the first network based on the compound error metric. The executable operations include updating weights of the second network using a moving average technique that is dependent on the weights of the first network as updated.

[0006] This Summary section is provided merely to introduce certain concepts and not to identify any key or essential features of the claimed subject matter. Other features of the inventive arrangements will be apparent from the accompanying drawings and from the following detailed description.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** illustrates an example computing environment that is suitable for training a machine learning (ML) model and executing an ML model at runtime.

[0008] FIGS. **2**A and **2**B, taken collectively, illustrate a domain generalization (DG) framework in accordance with one or more embodiments of the disclosed technology.

[0009] FIG. **3** illustrates certain operative features of the DG framework of FIGS. **1** and **2**.

[0010] FIG. **4** is a method illustrating certain operative features of the DG framework of FIGS. **1**, **2**, and **3** in accordance with one or more embodiments of the disclosed technology.

[0011] FIG. **5** illustrates a DG framework trained to perform inference in accordance with one or more embodiments of the disclosed technology.

DETAILED DESCRIPTION

[0012] While the disclosure concludes with claims defining novel features, it is believed that the various features described within this disclosure will be better understood from a consideration of the description in conjunction with the drawings. The process(es), machine(s), manufacture(s) and any variations thereof described herein are provided for purposes of illustration. Specific structural and functional details described within this disclosure are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the features described in virtually any appropriately detailed structure. Further, the terms and phrases used within this disclosure are not intended to be limiting, but rather to provide an understandable description of the features described.

[0013] This disclosure relates to domain generalization for machine learning (ML) models. In accordance with the inventive arrangements described within this disclosure, methods, systems, and computer program products are provided for training an ML model to learn an invariance representation of different domains having distribution shifts. In accordance with one or more embodiments described within this disclosure, the training framework provided uses the domain(s) themselves as augmentations of a model. Using the domains as or in place of augmentations, the inventive arrangements are capable of training an ML model to accurately classify objects across a plurality of different domains having different data distributions. The resulting ML model, trained as described herein, is capable of extracting semantic information to classify objects while

remaining insensitive to domain specific features. In this regard, domain generalization is differentiated from domain adaptation. Domain generalization refers to generalizing an ML model to generate accurate predictions given data items across a plurality of domains that may include unseen domains whereas domain adaptation refers to adapting an ML model trained for a first domain to make accurate predictions in a second, known domain.

[0014] For purposes of illustration, consider an ML model trained to recognize a given object or class of object using data from a selected domain. That ML model may be unable to reliably recognize the same object or class of object in images of different domains. As a simplified example, consider the case where the ML model is trained to recognize cars from photorealistic images of cars. The ML model may reliably detect cars in such photorealistic image data but perform poorly when presented with image data specifying cars from different domains. The ML model, for example, may be unable to reliably detect cars within sketch images or cartoon images, which represent different domains than photorealistic images.

[0015] The embodiments described within this disclosure may be applied in any of a variety of real-world contexts relating to image processing to process data, e.g., images, from multiple different domains. Such data, e.g., from different domains, is also referred to as "out-of-distribution" or "OOD" data. For purposes of illustration and not limitation, the embodiments may be used in the field of medical image processing. In that context, different domains of image data may include each different medical site being considered a different domain in consequence of differences in the equipment used to generate the images. For example, differences in scanners, protocols, patient cohorts, and the like taken individually or in any combination may result in different domains of images to which the training embodiments and inference embodiments described herein may be used. Whereas with conventional ML models and ML training techniques, differences in medical sites, scanners, protocols, patient cohorts, and the like may result in different image domains that cause the ML model to fail to correctly classify images (and thereby fail to make an accurate diagnosis), the inventive arrangements described herein provide the technical effect of more accurately and/or more reliably classifying images from such different domains. Appreciably, medical images are used for purposes of illustration. The inventive arrangements are not intended to be limited to any particular application or context.

[0016] In some aspects, the techniques described herein relate to training a ML model such that the ML model, as trained, is able to classify images across different domains. The techniques include processing, using computer hardware, a first plurality of images belonging to a first domain through a first network. For example, images of the first domain are processed by the first network only. This provides the technical effect that the first network learns features of images of the same class and also features that may be particular to the first domain. The techniques include processing, using the computer hardware, a second plurality of images belonging to a second domain through a second network. For example, images of the second domain are processed by the second network only. This provides the technical effect that the second network learns features of images of the same class and also features that may be particular to the second domain.

[0017] The techniques include generating, using the computer hardware, a compound error metric (L_{total}) from a plurality of error metrics (e.g., L_{mse}, L_{ce}, L_{SC}) derived from results generated from the processing of the first network and the processing of the second network. The use of a compound error metric provides the technical effect of addressing a plurality of different goals in the training process concurrently. For example, the compound error metric, as used during the training process, includes components (e.g., $L_{intraSC}$ and $L_{interSC}$) that ensure that the training process accounts for differences between different classes of objects whether such objects are members of a same domain or are members of different domains. The compound error metric, as used during the training process, includes components (e.g., $L_{interSC}$, L_{mse}) that ensure that the training process, for images of the same class and from different domains, learns or accounts for features particular to the class(es) while discounting features that are attributable to the domains.

[0018] The techniques include updating, using the computer hardware, weights of the first network based on the compound error metric. In one or more examples, the weights of the first network may be trained using a gradient descent technique based on the compound error metric. The technical effect of updating the first network with this technique is that the network updates more often and adapts faster to the images that are processed. The techniques also include updating, using the computer hardware, weights of the second network using a moving average technique that is dependent on the weights of the first network as updated. The second network is updated using a weight updating technique that is different from the weight updating technique of the first network. The technical effect of updating the second network with this technique is that the second network is updated more slowly based on a moving average of weights of the first network.

[0019] By using two networks that, as described hereinbelow, are architecturally similar or the same, but have different weights owing, at least in part to the update processes used, additional error metrics may be determined that are included in the compound error metric. The architecture disclosed herein provides technical effects similar to those of student-teacher networks or knowledge distillation. The target network is slowly updated and is able to learn from previously seen data, which helps the target network to remember such data (e.g., not to forget). The online network learns to approximate the predicted targets produced by the target network, which has the benefit of memory.

[0020] In some aspects, the first plurality of images and the second plurality of images are ordered according to class as processed by the first network and the second network. Ordering the images provided to the respective networks provides the technical effect that both networks are processing images of the same class albeit from different domains concurrently or simultaneously as the case may be. Thus, any error metrics that are calculated are able to distinguish between features attributable to the class of the images and features attributable to the domains of the image. These error metrics, as embodied in the compound error metric, influence the training and updating of weights in the respective networks. In one or more aspects, the error metrics as embodied in the compound error metric may directly influence weights of the first network and indirectly influence weights of the second network.

[0021] In some aspects, the first network includes an online encoder, an online projector, an online predictor, and a classifier. The second network includes a target encoder and a target projector. The first network, through inclusion of a classifier, facilitates the generation of an error metric (e.g., L_{ce}) relating to the operation of the classifier in predicting the classes or labels of the first plurality of images processed against the ground truth labels (e.g., classes) of respective ones of the first plurality of images.

[0022] In some aspects, the classifier is configured to receive embeddings generated by the online encoder. The technical effect of providing embeddings from the online encoder is that the embedding, as output from the online encoder, is configured to move the representation of the first plurality of images in the embeddings closer to those of the embeddings generated by the second network which processes the second plurality of images of the second domain. Thus, the classifier is learning to predict the class of received images from an embedding space in which images of same class are closer to one another than images of different classes regardless of domain membership.

[0023] In some aspects, the plurality of error metrics includes a selected error metric (e.g., L_{ce}) generated based on classification results from the classifier of the first network and ground truth labels of the first plurality of images. A technical effect of the selected error metric is that the performance of the classifier in predicting the labels of received images is generated with respect to a ground truth label. Thus, the compound error metric, as applied through training, updates the weights of the classifier to more accurately predict the label of received images even as images of different domains may be provided over time at least at inference time.

[0024] In some aspects, the plurality of error metrics include a supervised contrastive loss (e.g., L_{SC}). A technical effect of the supervised contrastive loss is that the error metric accounts for both similarities in images of the same class in the same domain and similarities between images of the same class in different domains. The supervised contrastive loss can include an intra-domain error (e.g., $L_{intraSC}$), e.g., a component, generated by comparing images in same mini-batches having same labels only for the first plurality of images. The technical effect of including the intra-domain error is the ability to specify or quantify a measure of error between embeddings of images in the same mini-batch, having same labels (e.g., being of the same class), and belonging to the same domain. The supervised contrastive loss can include an inter-domain error (e.g., $L_{interSC}$), e.g., another component, generated by comparing mini-batches of images of the first plurality of images with mini-batches of images of the second plurality of images, wherein the inter-domain error compares images with same labels. The technical effect of the inter-domain error is the ability to specify or quantify a measure of error between embeddings of images of the first domain and embeddings of images of the second domain, wherein the images have the same label (e.g., are of the same class). It should be appreciated that while comparisons of images are referenced, the comparisons and generation of the respective components of the supervised contrastive loss are performed using embeddings.

[0025] In some aspects, the plurality of error metrics includes a selected error metric (e.g., L_{mse}) generated based on a prediction similarity between the first network and the second network. The L_{mse} is used to measure similarity between features of the first network and features of the second network.

[0026] The predictor module is in the online network since it is updated using gradients and we are learning a classification task. We only use it for the online network since the division of domains to online and target comes from the field of meta-learning where we simulate a situation of an unseen domain in the target path and we don't want to learn the classification task on it.

[0027] The inventive arrangements provide additional technical effects in that a single or same ML model may be used across different domains. The ability to generalize an ML model across different domains means that the same ML model may be utilized to obtain reliable results for more than one domain rather than training and deploying a plurality of different ML models. Moreover, using a plurality of domain specific models requires that the domain of images be known a priori when this is not always knowable. By comparison, an ML model trained for domain generalization as described herein may process images of different domains without knowing a priori the domain(s) to which the images belong and may also operate on domains unseen during training.

[0028] Further aspects of the embodiments described within this disclosure are described in greater detail with reference to the figures below. For purposes of simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference numbers are repeated among the figures to indicate corresponding, analogous, or like features.

[0029] Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

[0030] A computer program product embodiment ("CPP embodiment" or "CPP") is a term used in the present disclosure to describe any set of one, or more, storage media (also called "mediums") collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A "storage device" is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash

memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

[0031] FIG. **1** illustrates a computing environment **100** that is suitable for training an ML model and executing an ML model (e.g., at runtime or inference time). Referring to FIG. **1**, computing environment **100** contains an example of an environment for the execution of at least some of the computer code in block **150** involved in performing the inventive methods, such as domain generalization (DG) framework **200** implemented as executable program code or instructions. DG framework **200** is capable of training an ML model to reliably recognize or classify objects at runtime that belong to one or more different domains and potentially domains that were not seen during training. Within this disclosure, the term "runtime" or "inference time," in the context of an ML model, means the time that the ML model is run or executed by computer hardware post training to perform inference. Within this disclosure, the term "domain" means a data set having a data distribution that differs from the data distribution of another, different data set. Thus, two domains including objects belonging to one or more same classes will have different data distributions. The resulting ML model, as trained using DG framework **200**, is capable of reliably classifying received input across the different domains. The inventive arrangements, as embodied in DG framework **200**, are capable of performing executable operations such as those described herein in connection with FIGS. **2**, **3**, **4**, and **5**.

[0032] In addition to block **150**, computing environment **100** includes, for example, computer **101**, wide area network (WAN) **102**, end user device (EUD) **103**, remote server **104**, public cloud **105**, and private cloud **106**. In this embodiment, computer **101** includes processor set **110** (including processing circuitry **120** and cache **121**), communication fabric **111**, volatile memory **112**, persistent storage **113** (including operating system **122** and block **150**, as identified above), peripheral device set **114** (including user interface (UI) device set **123**, storage **124**, and Internet of Things (IoT) sensor set **125**), and network module **115**. Remote server **104** includes remote database **130**. Public cloud **105** includes gateway **140**, cloud orchestration module **141**, host physical machine set **142**, virtual machine set **143**, and container set **144**.

[0033] COMPUTER **101** may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database **130**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment **100**, detailed discussion is focused on a single computer, specifically computer **101**, to keep the presentation as simple as possible. Computer **101** may be located in a cloud, even though it is not shown in a cloud in FIG. **1**. On the other hand, computer **101** is not required to be in a cloud except to any extent as may be affirmatively indicated.

[0034] PROCESSOR SET **110** includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry **120** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **120** may implement multiple processor threads and/or multiple processor cores. Cache **121** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **110**. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located "off chip." In some computing environments, processor set **110** may be designed for working with qubits and performing quantum computing.

[0035] Computer readable program instructions are typically loaded onto computer **101** to cause a series of operational steps to be performed by processor set **110** of computer **101** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as "the inventive methods"). These computer readable program instructions are stored in various types of computer readable storage media, such as cache **121** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **110** to control and direct performance of the inventive methods. In computing environment **100**, at least some of the instructions for performing the inventive methods may be stored in block **150** in persistent storage **113**.

[0036] COMMUNICATION FABRIC **111** is the signal conduction paths that allow the various components of computer **101** to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless communication paths.

[0037] VOLATILE MEMORY **112** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, the volatile memory is characterized by random access, but this is not required unless affirmatively indicated. In computer **101**, the volatile memory **112** is located in a single package and is internal to computer **101**, but, alternatively or additionally, the volatile

memory may be distributed over multiple packages and/or located externally with respect to computer **101**.

[0038] PERSISTENT STORAGE **113** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **101** and/or directly to persistent storage **113**. Persistent storage **113** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid-state storage devices. Operating system **122** may take several forms, such as various known proprietary operating systems or open-source Portable Operating System Interface type operating systems that employ a kernel. The code included in block **150** typically includes at least some of the computer code involved in performing the inventive methods.

[0039] PERIPHERAL DEVICE SET **114** includes the set of peripheral devices of computer **101**. Data communication connections between the peripheral devices and the other components of computer **101** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion type connections (e.g., secure digital (SD) card), connections made though local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **123** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **124** may be persistent and/or volatile. In some embodiments, storage **124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **101** is required to have a large amount of storage (e.g., where computer **101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

[0040] NETWORK MODULE **115** is the collection of computer software, hardware, and firmware that allows computer **101** to communicate with other computers through WAN **102**. Network module **115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **115** are performed on the same physical hardware device. In other embodiments (e.g., embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **101** from an external computer or external storage device through a network adapter card or network interface included in network module **115**.

[0041] WAN **102** is any wide area network (e.g., the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now known or to be developed in the future. In some embodiments, the WAN may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

[0042] END USER DEVICE (EUD) **103** is any computer system that is used and controlled by an end user (e.g., a customer of an enterprise that operates computer **101**), and may take any of the forms discussed above in connection with computer **101**. EUD **103** typically receives helpful and useful data from the operations of computer **101**. For example, in a hypothetical case where computer **101** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **115** of computer **101** through WAN **102** to EUD **103**. In this way, EUD **103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **103** may be a client device, such as a thin client, heavy client, mainframe computer, desktop computer and so on.

[0043] REMOTE SERVER **104** is any computer system that serves at least some data and/or functionality to computer **101**. Remote server **104** may be controlled and used by the same entity that operates computer **101**. Remote server **104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **101**. For example, in a hypothetical case where computer **101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **101** from remote database **130** of remote server **104**.

[0044] PUBLIC CLOUD **105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economics of scale. The direct and active management of the computing resources of public cloud **105** is performed by the computer hardware and/or software of cloud orchestration module **141**. The computing resources provided by public cloud **105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **142**, which is the universe of physical computers in and/or available to public cloud **105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **143** and/or containers from container set **144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **141** manages the transfer

and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **140** is the collection of computer software, hardware, and firmware that allows public cloud **105** to communicate through WAN **102**.

[0045] Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as "images." A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

[0046] PRIVATE CLOUD **106** is similar to public cloud **105**, except that the computing resources are only available for use by a single enterprise. While private cloud **106** is depicted as being in communication with WAN **102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (e.g., private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **105** and private cloud **106** are both part of a larger hybrid cloud.

[0047] FIGS. **2**A and **2**B, taken collectively and collectively referred to as FIG. **2**, illustrate DG framework **200** in accordance with one or more embodiments of the disclosed technology. In one or more embodiments, DG framework is capable of operating as a domain invariant feature extractor. DG framework **200** is capable of training an ML model to capture semantic information while remaining insensitive to domain specific features. In one or more embodiments, DG framework **200** incorporates one or more features of a self-supervised learning framework as generally described in Grill et al., "Bootstrap Your Own Latent-A New Approach to Self-Supervised Learning," Advances in Neural Information Processing Systems, v. 33, pp. 21271-21284, Curran Associates, Inc. (2020) (hereafter BYOL).

[0048] In the example of FIG. **2**, DG framework **200** includes an online network **210** and a target network **220**. In one or more embodiments, online network **210** and target network **220** may be implemented as convolutional neural networks. Online network **210** is also referred to herein as the "first network." Target network **220** is also referred to herein as the "second network." In the example, online network **210** includes an online encoder **212** (f_θ), an online projector **214** (g_θ), an online predictor **216** (q_θ), and a classifier **218** (c_θ). Target network **220** includes a target encoder **222** (f_ξ) and a classifier **218** (g_ξ). For purposes of illustration, within this disclosure, parameters of online network **210** are referred to as e while the parameters of target network **220** are referred to as ξ.

[0049] In one or more embodiments, online predictor **216** is included in online network **210** as online network **210**, inclusive of online predictor **216**, is updated using gradients and a classification task is being learned. Online predictor **216** is included in online network **210** as DG framework **200**, for training purposes, simulates the situation of an unseen domain in target network **220** where a classification task is not being learned.

[0050] In accordance with the inventive arrangements, and unlike the self-supervised learning framework described in BYOL, online network **210** receives images **202** of one or more classes that belong to a first domain also referred to herein as D_i. Target network **220** receives images **204** of the same one or more classes. Whereas images **202** belong to a first domain D_i, images **204** belong to a second domain D_j. Images **202** may be provided to online network **210** and images **204** provided to target network **220** in mini-batch sizes of N images. Each image may be provided to online network **210** and to target network **220** as an image-label pair denoted as (x_k, y_k), where k=1, . . . , N. In the example, domain D_i is part of a meta-train set that is used in online network **210**. Domain D_j is part of a meta-test set used in target network **220**. Within this disclosure, though the term "label" refers to a data item that specifies a class, the terms "class" and "label" may be used interchangeably from time-to-time.

[0051] With the exception of online predictor **216** and classifier **218**, online network **210** and target network **220** are substantially similar to one another. Online encoder **212** and online projector **214** may be implemented, from an architectural perspective, to be the same as target encoder **222** and target projector **224**, respectively. That is, though online encoder **212** and target encoder **222** may be architecturally the same, each encoder will have different weights that are updated during training using different weight updating techniques. Similarly, though online projector **214** and target projector **224** may be architecturally the same, each projector will have different weights that are updated during training using different weight updating techniques. Thus, the weights of online network **210** are updated differently, e.g., using a different weight updating technique, than the weights of target network **220**.

[0052] DG framework **200** includes additional blocks such as a classifier **218**, a plurality of error calculators, and a loss calculator **260**. The error calculators include a cross-entropy calculator **240**, an intra-domain error calculator **242**, an inter-domain error calculator **244**, and a BYOL regressor **246**. Cross-entropy calculator **240** is capable of generating an error metric such as a cross-entropy error **250**. Intra-domain error calculator **242** is capable of generating an error metric such as an intra-domain error **252**. Inter-domain error calculator **244** is capable of generating an error metric such as an inter-domain error **254**. BYOL regressor **246** is capable of generating an error metric such as a BYOL regression error **256**.

[0053] Supervised contrastive loss calculator **248** receives intra-domain error **252** and inter-domain error **254**. Based on intra-domain error **252** and inter-domain error **254**, supervised contrastive loss calculator **248** generates supervised contrastive loss **258**. As illustrated, loss calculator **260** is capable of receiving the respective metrics. For example,

loss calculator **260** receives cross-entropy error **250**, supervised contrastive loss **258**, and BYOL regression error **256**. Based on the received metrics, loss calculator **260** generates a compound error metric **262**. Online network **210** is trained using, at least in part, based on compound error metric **262**.

[0054] In the example, online network **210** generates, e.g., outputs a representation vector. Similarly, target network **220** generates, e.g., outputs, a representation vector. The training process is operative to bring the two representation vectors, corresponding to two different domains, close together. For example, BYOL regressor **246** is capable of comparing these representation vectors and outputting the error metric BYOL regression error **256**. Further aspects of DG framework **200** are described in greater detail below in connection with FIGS. **3** and **4**.

[0055] FIG. **3** illustrates certain operative features of DG framework **200** of FIGS. **1** and **2**. In the example of FIG. **3**, DG framework **200** is trained using a plurality of domains **302**. In the example, domains D_i and D_j are illustrated with domain D_i including a plurality of images **202-1**, **202-2**, and **202-3** and domain D_j including a plurality of images **204-1**, **204-2**, and **204-3**. For purposes of illustration, domain D_i includes photorealistic images such as images taken with a camera. Domain D_j includes images that are cartoons. Each domain D_i and D_j includes images of the same set of classes. For example, image **202-1** and image **204-1** correspond to class 1, image **202-2** and image **204-2** correspond to class 2, and image **202-3** and image **204-3** correspond to class 3. For purposes of discussion in this example, class 1 is formed of images that include a guitar. Class 2 is formed of images that include a horse. Class 3 is formed of images that include a house.

[0056] In providing a mini-batch of N images **202** to online network **210** and a mini-batch of N images **204** to target network **220**, images **202** and **204** are aligned according to class (e.g., label). For example, online network **210** and target network **220** each receives a mini-batch of images where the images of both mini-batches belong to the same class. Online network **210** and target network **220**, for example, may process mini-batches of images until the images of that class are complete, followed by one or more mini-batches of images of a next class, etc. Further, during training, online network **210** and target network **220** receive images of the same class but from different domains. For example, online network **210** receives an image **202**, from domain D_i, with a label of class 1 concurrently or simultaneously with target network **220** receiving an image **204** from domain D_j, also with a label of class 1.

[0057] Output **304** illustrates that DG framework **200** is trained to bring together images having same classifications whether the images are within the same domain or exist in different domains. DG framework **200** is further trained to drive apart images of different classifications whether such images are in the same domain or in different domains. In the example, solid arrows with arrowheads facing outward represent repulsive forces pushing images apart in an embedding space (e.g., the embedding space(s) as output from the encoders and/or as output from the projectors) while dashed arrow with the arrowheads pointing in toward one another represent attractive forces that pull images closer to one another in the embedding space (e.g., as measured using a distance metric). As illustrated, image pairs (**202-1**, **204-1**) both labeled as "class 1," (**202-2**, **204-2**) both being labeled "class 2," and (**202-3**, **204-3**) both being labeled "class 3" are joined by an attractive force that brings the respective pairs of images closer to one another in the embedding space. Conversely, images pairs (**202-1**, **204-3**), (**204-1**, **204-3**), (**204-3**, **202-2**), and (**204-3**, **204-2**) are coupled by repulsive forces as the images in each pair are labeled with different classes and are pushed apart in the embedding space.

[0058] Example 1 below is pseudo code illustrating operations performed by DG framework **200** of FIGS. **1** and **2** in training the ML model. For purposes of illustration, the trained ML model is formed of online encoder **212** and classifier **218**, which is illustrated as the inference system in FIG. **5**.

Example 1

1. Input : Domains S, Model θ, U # meta-test
2. Output : L_{total}
3. for itr in iterations do
4. Split S domains to $\{D_i | i = S - U\}$, $\{D_j | j = 1...U\}$
5. $x_i, y_i, x_j, y_j \leftarrow$ OrderDomains(x_i, y_i, x_j, y_j)
6. $p_i, z_j, y_i, u_i, u_j \leftarrow$ BYOLDG(x_i, x_j)
7. $L_{CE} \leftarrow$ CrossEntropy(xi, yi)
8. $L_{MSE} \leftarrow$ MSE(p_i, z_j)
9. $L_{SC} \leftarrow$ intraSC(u_i, u_i) + interSC(u_i, u_j)
10. Optimize Update $\theta \leftarrow$ Adam($\alpha L_{CE} + \beta L_{MSE} + \gamma L_{SC}$)
11. end

[0059] In one or more embodiments, in reference to Example 1, inputs to online network **210** and target network **220** may be switched to simulate evaluation of the meta-test as an "unseen" domain so that only x_i undergoes classification. The model is trained on all pairs of S, U in aggregate.

[0060] FIG. **4** is a method **400** illustrating certain operative features of DG framework **200** of FIGS. **1**, **2**, and **3** in accordance with one or more embodiments of the disclosed technology. Method **400** illustrates example operations performed by DG framework **200** in training an ML model to reliably or accurately classify objects in images across a plurality of domains (e.g., train a domain generalized ML model). For example, the operations illustrated in FIG. **4** correspond to the operations of the loop body of Example 1 corresponding to lines **4-10**.

[0061] In the context of FIG. **4**, there are a set of M training domains called "source domains" denoted as $S_{train}=\{D_m | m=1, \ldots, M\}$, where $D_m=\{x_i, y_i\}_{i=1}^{N_m} \sim P_{XY}^{(m)}$ is m-th domain with N_m data samples $x_i \in X \subset \mathbb{R}^d$, their corresponding labels $y_i \in Y \subset \mathbb{R}$ and $P_{XY}=P(X, Y)$ is the joint distribution of input samples and labels, where X and Y denote the corresponding random variables. The joint distributions of the domains are pairwise different: $P_{XY}^{(m)} \neq P_{XY}^{(m')}$ for all $m \neq m'$. In general, the ML model is trained to learn a function f: $X \rightarrow Y$ from training domains that minimizes prediction error on a test domain S_{test} that is different from all of the source domains.

[0062] In some embodiments, the ML model may be considered to be trained where or when a loss function (e.g., the compound error metric **262**) is minimized and/or upon the accuracy of the trained ML model being greater than or equal to a threshold accuracy score (e.g., 85% or greater, 90% or greater, 95% or greater, etc.) obtained using test data. In some embodiments, some or all of the weights and biases learned by the ML model may be retained for inference operations for downstream tasks.

[0063] In block **402**, the domains are split for processing. Block **402** corresponds to line **4** of Example 1 above. In the example of FIG. **4**, a meta-learning domain selection scheme is used for each mini-batch. The splitting operation of block **402** may be performed as follows: given S training domains $\{D_k\}_{k=1,\ldots,s}$, the domains S are randomly split into S-U meta-train domains $\hat{S}$ with input pairs $\{x_i, y_i\}_{i=1}^{n}$ and U meta-test domains with $\{x_j, y_j\}_{j=1}^{m}$. The meta-train set $\hat{S}$ follows, e.g., is provided as input to, online network **210** while the meta-test set U follows, e.g., is provided as input to, target network **220**. The number of meta test domains U is a hyper-parameter.

[0064] In block **404**, the domains are ordered according to class. Block **404** corresponds to line **5** of Example 1 above. For example, the images of each domain are ordered according to class or label. The ordering ensures that as pairs of images are provided to DG framework **200**, a first image of each pair is provided to online network **210** and the second image of each pair is provided to target network **220**. For each pair of images provided to DG framework **200**, both images are of the same class and, as such, have the same label, but belong to different domains.

[0065] In block **406**, DG framework **200** processes images. Image pairs are processed through online network **210** and through target network **220**. Block **406** corresponds to line **6** of Example 1 above. Images are provided to DG framework **200** in pairs as described where online network **210** receives the first image of each pair while target network **220** receives the second image of each pair.

[0066] For example, in block **408**, a first plurality of images (e.g., a mini-batch of images N), is processed through online network **210**. As noted, online network **210** is also referred to as the "first network." The first plurality of images belong to a first domain. Each image of the first plurality of images is processed through online encoder **212** to obtain an embedding u, which is provided to classifier **218** and to online projector **214**. Classifier **218**, based on the received embedding u, generates a classification result $\hat{y}$ for each image of the first plurality of images. Online projector **214**, based on the embedding u, generates an output z. Online predictor **216**, based on the output z from online projector **214**, outputs a prediction denoted as $q_\theta(z)$.

[0067] In general, online projector **214** projects embeddings u into a different size corresponding to z. Typically, online projector **214** has the effect of reducing the number of features significantly. In addition to the reduction in feature size, which makes the embeddings more manageable and reduces the computational overhead of DG framework **200**, the embeddings u are further projected into embedding space z that is closer to the embedding space $\hat{z}$ corresponding to target network **220**. Target projector **224** operates substantially similar to online projector **214** albeit having different weights as previously noted. Accordingly, target projector **224** projects embeddings $\hat{u}$ into a different size corresponding to $\hat{z}$. Typically, target projector **224** has the effect of reducing the number of features significantly. In addition to the reduction in feature size, which makes the embeddings more manageable and reduces the computational overhead of DG framework **200**, the embeddings $\hat{u}$ are further projected into the embedding space $\hat{z}$ that is closer to the space z corresponding to online network **210**.

[0068] In block **410**, a second plurality of images (e.g., a mini-batch of images N) is processed by target network **220**. As noted, target network **220** is also referred to as the "second network." The second plurality of images belongs to a second domain (e.g., a domain different from the domain of the first plurality of images). Each image of the second plurality of images is processed through target encoder **222** to obtain an embedding $\hat{u}$, which is provided to target projector **224**. Target projector **224**, based on the embedding $\hat{u}$, generates an output **2**.

[0069] In one or more embodiments, the class of the images provided to DG framework **200** in pairs may be held constant over each mini-batch of N images. Once a mini-batch is processed, the class of image pairs for the next mini-batch may be changed to a different class. In another aspect, the class of image pairs for the next mini-batch and/or further mini-batches may continue or remain unchanged until all images of that class for the two domains have been exhausted, at which point image pairs belonging to different domains for a different class may be fed into DG framework **200**.

[0070] In block **412**, DG framework **200** generates compound error metric **262** from a plurality of error metrics. The error metrics used to generate compound error metric **262** are derived from results generated from the processing of the first network and the processing of the second network. For example, in block **414**, DG framework **200** generates an error metric based on a predicted similarity between online network **210** and target network **220**. Block **414** corresponds to line **8** of Example 1 above. More particularly, in block **414**, BYOL regressor **246** generates the BYOL regression error **256**. The BYOL regression error **256** is shown below in Expression 1 as L_{mse}. As discussed, q_θ represents online predictor **216**. The terms z and $\hat{z}$ are the outputs of the of online projector **214** and target projector **224**, respectively.

$$L_{mse} = 2 - 2 \cdot \frac{\langle q_\theta(z), \hat{z} \rangle}{\|q_\theta(z)\|_2 \cdot \|\hat{z}\|_2}. \quad (1)$$

[0071] In Expression 1, BYOL regressor **246** calculates the mean squares error as a loss function. In the case of BYOL, the error pushes the ML model to learn the augmentation that was applied to the image and demand the prediction similarity between the images (image and augmented image) independent of the augmentation applied. In the case of DG framework **200**, the relationship between the two domains corresponding to online network **210** and target network **220** is viewed or considered a similarity problem. Thus, BYOL regression error **256** in the context of DG framework **200** pushes the ML model to learn the transformation between the two domains being processed for the mini-batch of images.

[0072] In block **416**, DG framework **200** generates an error metric based on a classification result and ground truth label referred to herein as cross-entropy error **250**. Block **416** corresponds to line **7** of Example 1 above. For example, cross-entropy error calculator **240** calculates a cross entropy error **250**, which is also denoted herein as L_{CE}. The cross-entropy error **250** is calculated based on the classification results from classifier **218** and ground truth labels (e.g., y_i) of the first plurality of images and is shown below in Expression 2.

[0073] Referring to FIG. **2**, the augmentations applied to images are implemented or realized as the inherent difference between domains. DG framework **200** solves a supervised task and uses the labels to match between domain

images to encourage domain similarity. The set of images and labels is defined as (x_i, y_i) for online network **210** and (x_j, y_j) for target network **220**. As discussed, each mini-batch is ordered according to the image labels such that the target images (e.g., images **204**) and the online images (e.g., image **202**) are paired with the same class. The embeddings u obtained from online encoder **212** are fed into classifier **218**. Cross-entropy calculator **240** is capable of generating cross-entropy error **250** using Expression 2. Within Expression 2 below, $\hat{y}_i$ is the predicted label and y_i is the ground truth label.

$$L_{ce} = -\frac{1}{N}\sum_{i=0}^{N} y_i \log(\hat{y}_i) \tag{2}$$

[0074] In block **418**, DG framework **200** generates an error metric referred to as supervised contrastive loss **258**. Block **418** corresponds to line **9** of Example 1 above. Supervised contrastive loss **258** is also denoted herein as L_{SC}. As illustrated, supervised contrastive loss calculator **248** generates supervised contrastive loss **258** based on two different components. The first component is intra-domain error **252**, which is also denoted as $L_{intraSC}$ within this disclosure. In general, intra-domain error **252** is generated by comparing images in same mini-batches having same labels only for the first plurality of images (e.g., for online network **210**). The second component is inter-domain error **254**, which is also denoted as $L_{interSC}$ within this disclosure. In general, inter-domain error **254** is generated by comparing mini-batches of images of the first plurality of images with mini-batches of images of the second plurality of images. Still, the comparisons performed for inter-domain error **254** compare images with same labels.

[0075] Intra-domain error **252** and inter-domain error **254** ensure that inter-domain representation differences are minimized while intra-domain variations are also considered. Thus, supervised contrastive loss **258** ensures that all sample representations from the same class in a mini-batch, whether such sample representations are from the same domain or different domains, are brought closer together in the embedding space.

[0076] Referring to intra-domain error **252** as generated by intra-domain error calculator **242**, a mini-batch of N samples is defined as $\{x_k, y_k\}_{k=1,\ldots,N}$ with representation vectors $\{u_k\}_{k=0}^{N}$ as the output of online encoder **212**. Intra-domain error **252** is computed over a mini-batch and compares each example in the mini-batch to the other examples in the mini-batch that share the same label. For each k, $P(k)=\{u_l | y_l=y_k, l\neq k\}$ is defined which includes all examples sharing the same label as u_k. Intra-domain error **252**, e.g., $L_{intraSC}$, is generated by intra-domain error calculator **242** using Expression 3 below. Within Expression 3, τ is a temperature hyperparameter.

$$L_{intraSC} = \sum_{k\in N}\frac{-1}{|P(k)|}\sum_{u_1\in P(k)}\log\frac{\exp\left(u_k\cdot\frac{u_l}{\tau}\right)}{\sum_{\substack{n\in N\\ n\neq k}}\exp\left(u_k\cdot\frac{u_n}{\tau}\right)} \tag{3}$$

[0077] Inter-domain error **254**, as generated by inter-domain error calculator **244**, is similar to intra-domain error **252**. Instead of comparing a mini-batch to itself, inter-domain error calculator **244** compares examples of a mini-batch of one domain D_i to the examples of another domain D_i with representations $\{u_k\}_{k=0}^{N}$ and $\{\hat{u}_k\}_{k=0}^{N}$, respectively. In the case of inter-domain error **252**, $\hat{P}(k)=\{\hat{u}_l | \hat{y}_l=y_k\}$ is defined. Inter-domain error **254**, e.g., $L_{interSC}$, is generated by inter-domain error calculator **244** using Expression 4 below.

$$L_{interSC} = \sum_{k\in N}\frac{-1}{|P(k)|}\sum_{\hat{u}_1\in\hat{P}(k)}\log\frac{\exp\left(u_k\cdot\frac{\hat{u}_l}{\tau}\right)}{\sum_{n\in N}\exp\left(u_k\cdot\frac{\hat{u}_n}{\tau}\right)} \tag{4}$$

[0078] As part of block **418**, supervised contrastive loss calculator **248** generates supervised contrastive loss **258** by summing intra-domain error **252** and inter-domain error **254** in accordance with Expression 5 below.

$$L_{SC} = L_{intraSC} + L_{interSC} \tag{5}$$

[0079] In block **420**, DG framework **200** and, more particularly, loss calculator **260**, generates a weighted sum of the error metrics as compound error metric **262** in accordance with Expression 6 below where α, β, and γ are weights for the respective error metrics. Compound error metric **262** is also denoted herein as L_{total}.

$$L_{total} = \alpha L_{CE} + \beta L_{MSE} + \gamma L_{SC} \tag{6}$$

[0080] In block **422**, the weights of online network **210** are updated. In one or more embodiments, the weights of online network **210** are updated using a gradient descent technique based on compound error metric **262**.

[0081] In block **424**, the weights of target network **220** are updated. In one or more embodiments, the weights of target network **220** are updated using a moving average technique. The moving average technique is dependent on, e.g., is a moving average of, the weights of online network **210**. In one or more embodiments, the moving average technique is an exponential moving average technique. The updating of weights of target network **220** is an example of a momentum updating scheme.

[0082] In one or more embodiments, during the training process, online network **210** is updated at every step using the optimizer rule in reference to block **422**. In reference to block **424**, target network **220** may be updated using Expression 7 below. In Expression 7, τ is a weight decay parameter in the range of [0, 1] and s is the update step.

$$\xi_s = \tau\xi_{s-1} + (1-\tau)\theta \tag{7}$$

[0083] Thus, in reference to blocks **422** and **424**, in each mini-batch, DG framework **200** computes the compound error metric **262**. This loss is backpropagated to update the weights of online network **210** and, more particularly, online encoder **212**, online projector **214**, online predictor **216**, and classifier **218**. The gradient of the respective components is

calculated and the components (e.g., weights thereof) updated. The weights of target network **220** and, more particularly, target encoder **222** and target projector **224** follow the weights of online network **210** as a moving average of those weights. Blocks **420**, **422**, and **424** generally correspond to line **10** of Example 1 above.

[0084] As discussed, method **400** illustrates operations performed as part of the body of the loop illustrated in Example 1. As noted, in one or more embodiments, inputs to online network **210** and target network **220** may be switched to simulate evaluation of the meta-test as an "unseen" domain so that only x_t undergoes classification. The model is trained on all pairs of $\hat{S}$, U in aggregate. The training process described herein may continue or iterate for a predetermined number of iterations or until the ML model converges to a particular desired level of accuracy.

[0085] At runtime, e.g., inference time, the only components retained for operation are online encoder **212** and classifier **218**. The other components illustrated in FIG. **2** are discarded.

[0086] FIG. **5** illustrates a DG framework **500** trained to perform inference in accordance with one or more embodiments of the disclosed technology. As shown, DG framework **500** includes online encoder **212** and classifier **218** as trained per the embodiments described herein. DG framework **500** is capable of accurately classifying received images, e.g., image **502**, from one or more different domains. DG framework **500** is capable of outputting a classification result **504** such as a predicted label specifying a particular class for image **502**.

[0087] The inventive arrangements described herein provide methods, systems, and computer-program products that implement domain generalization by using the inherent difference between domains as a transformation to be learned by bringing together similar object representations from different domains. In accordance with the inventive arrangements, two images are given as input to the DG framework instead of a single image with two augmentations. A mean squares error may be used as a loss function where to learn the augmentation that was applied to the image and demand the prediction similarity between the images independent of the transformation. In accordance with the inventive arrangements, the inherent difference between domains is considered instead of the augmentations applied in the self-supervised case. The DG framework learns to solve a supervised task with Cross-Entropy loss and using the labels to match between domain images in order to encourage domain similarity. A supervised contrastive loss is used where sample representations from the same class in a mini-batch and/or batch are brought closer together. Meta-learning may be used to train the DG framework, at least in part, by iteratively changing domain roles between meta-train and meta-test. As noted, domains are used as input as opposed to a single image, together with different loss combinations to give/provide improved domain invariance.

[0088] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. Notwithstanding, several definitions that apply throughout this document now will be presented.

[0089] As defined herein, the terms "at least one," "one or more," and "and/or," are open-ended expressions that are both conjunctive and disjunctive in operation unless explicitly stated otherwise. For example, each of the expressions "at least one of A, B and C," "at least one of A, B, or C," "one or more of A, B, and C," "one or more of A, B, or C," and "A, B, and/or C" means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

[0090] As defined herein, the term "automatically" means without user intervention.

[0091] As defined herein, the terms "includes," "including," "comprises," and/or "comprising," specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0092] As defined herein, the term "if" means "when" or "upon" or "in response to" or "responsive to," depending upon the context. Thus, the phrase "if it is determined" or "if [a stated condition or event] is detected" may be construed to mean "upon determining" or "in response to determining" or "upon detecting [the stated condition or event]" or "in response to detecting [the stated condition or event]" or "responsive to detecting [the stated condition or event]" depending on the context.

[0093] As defined herein, the term "responsive to" means responding or reacting readily to an action or event. Thus, if a second action is performed "responsive to" a first action, there is a causal relationship between an occurrence of the first action and an occurrence of the second action. The term "responsive to" indicates the causal relationship.

[0094] As defined herein, the terms "one embodiment," "an embodiment," "in one or more embodiments," "in particular embodiments," or similar language mean that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment described within this disclosure. Thus, appearances of the aforementioned phrases and/or similar language throughout this disclosure may, but do not necessarily, all refer to the same embodiment.

[0095] As defined herein, the term "output" means storing in physical memory elements, e.g., devices, writing to display or other peripheral output device, sending or transmitting to another system, exporting, or the like.

[0096] The terms "processor set" and "processor circuitry" refer to at least one hardware circuit configured to carry out instructions. The instructions may be contained in program code. The hardware circuit may be an integrated circuit. Examples include, but are not limited to, a central processing unit (CPU), an array processor, a vector processor, a digital signal processor (DSP), a field-programmable gate array (FPGA), a programmable logic array (PLA), an application specific integrated circuit (ASIC), programmable logic circuitry, and a controller.

[0097] The term "substantially" means that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations, and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.

[0098] The terms first, second, etc. may be used herein to describe various elements. These elements should not be limited by these terms, as these terms are only used to distinguish one element from another unless stated otherwise or the context clearly indicates otherwise.

[0099] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A method of training a machine learning model, the method comprising:

processing, using computer hardware, a first plurality of images belonging to a first domain through a first network;

processing, using the computer hardware, a second plurality of images belonging to a second domain through a second network;

generating, using the computer hardware, a compound error metric from a plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network;

updating, using the computer hardware, weights of the first network based on the compound error metric; and

updating, using the computer hardware, weights of the second network using a moving average technique that is dependent on the weights of the first network as updated.

2. The method of claim **1**, wherein the first plurality of images and the second plurality of images are ordered according to class as processed by the first network and the second network.

3. The method of claim **1**, wherein:

the first network includes an online encoder, an online projector, an online predictor, and a classifier; and

the second network includes a target encoder and a target projector.

4. The method of claim **3**, wherein the classifier is configured to receive embeddings generated by the online encoder.

5. The method of claim **1**, wherein the plurality of error metrics includes a selected error metric generated based on classification results from a classifier of the first network and ground truth labels of the first plurality of images.

6. The method of claim **1**, wherein the plurality of error metrics include a supervised contrastive loss comprising:

an intra-domain error generated by comparing images in same mini-batches having same labels only for the first plurality of images; and

an inter-domain error generated by comparing mini-batches of images of the first plurality of images with mini-batches of images of the second plurality of images, wherein the inter-domain error compares images with same labels.

7. The method of claim **1**, wherein the plurality of error metrics includes a selected error metric generated based on a prediction similarity between the first network and the second network.

8. A system for training a machine learning model, comprising:

one or more processors configured to execute operations including:

processing a first plurality of images belonging to a first domain through a first network;

processing a second plurality of images belonging to a second domain through a second network;

generating a compound error metric from a plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network;

updating weights of the first network based on the compound error metric; and

updating weights of the second network using a moving average technique that is dependent on the weights of the first network as updated.

9. The system of claim **8**, wherein the first plurality of images and the second plurality of images are ordered according to class as processed by the first network and the second network.

10. The system of claim **8**, wherein:

the first network includes an online encoder, an online projector, an online predictor, and a classifier; and

the second network includes a target encoder and a target projector.

11. The system of claim **10**, wherein the classifier is configured to receive embeddings generated by the online encoder.

12. The system of claim **8**, wherein the plurality of error metrics includes a selected error metric generated based on classification results from a classifier of the first network and ground truth labels of the first plurality of images.

13. The system of claim **8**, wherein the plurality of error metrics include a supervised contrastive loss comprising:

an intra-domain error generated by comparing images in same mini-batches having same labels only for the first plurality of images; and

an inter-domain error generated by comparing mini-batches of images of the first plurality of images with mini-batches of images of the second plurality of images, wherein the inter-domain error compares images with same labels.

14. The system of claim **8**, wherein the plurality of error metrics includes a selected error metric generated based on a prediction similarity between the first network and the second network.

15. A computer program product comprising one or more computer readable storage mediums having program instructions embodied therewith, wherein the program instructions are executable by one or more processors to cause the one or more processors to execute operations comprising:

processing a first plurality of images belonging to a first domain through a first network;

processing a second plurality of images belonging to a second domain through a second network;

generating a compound error metric from a plurality of error metrics derived from results generated from the processing of the first network and the processing of the second network;

updating weights of the first network based on the compound error metric; and

updating weights of the second network using a moving average technique that is dependent on the weights of the first network as updated.

16. The computer program product of claim **15**, wherein the first plurality of images and the second plurality of images are ordered according to class as processed by the first network and the second network.

17. The computer program product of claim **15**, wherein:

the first network includes an online encoder, an online projector, an online predictor, and a classifier; and

the second network includes a target encoder and a target projector.

18. The computer program product of claim **15**, wherein the plurality of error metrics includes a selected error metric generated based on classification results from a classifier of the first network and ground truth labels of the first plurality of images.

19. The computer program product of claim **15**, wherein the plurality of error metrics include a supervised contrastive loss comprising:

an intra-domain error generated by comparing images in same mini-batches having same labels only for the first plurality of images; and

an inter-domain error generated by comparing mini-batches of images of the first plurality of images with mini-batches of images of the second plurality of images, wherein the inter-domain error compares images with same labels.

20. The computer program product of claim **15**, wherein the plurality of error metrics includes a selected error metric generated based on a prediction similarity between the first network and the second network.

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